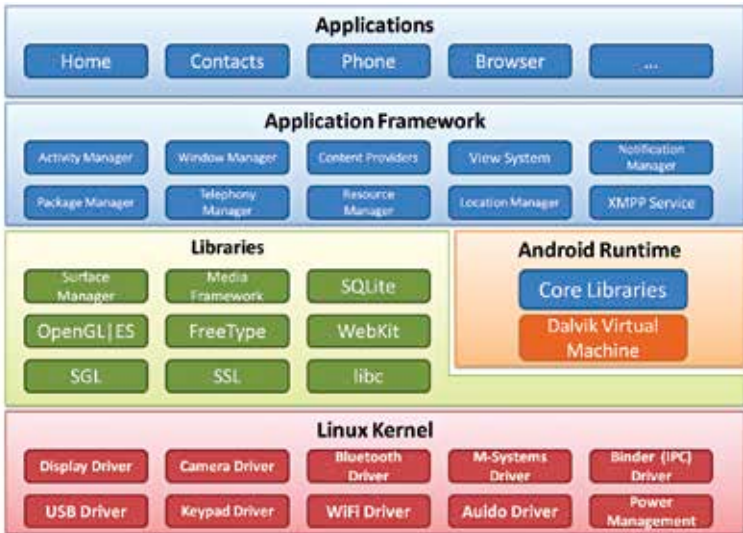


Protocols and stacks

The word protocol only means set of rules. The word does not convey anything more than that. A networking protocol is a set of rules for networking while a bus protocol would mean the set of rules for using the bus. An SoC is a complex hardware setup in a compact form-factor and normally has a standard bus governing the data transfer.

All peripherals use the bus. For the OS to orchestrate communication between multiple devices on the SoC, the devices and the OS must know how to operate on the bus. In addition, there might be other protocols of data transfer for each device. Also, there can be multiple buses in a system. For example, the bus used to transfer data from the RAM to the processor



Android Environment (the runtime can be either Dalvik or ART, depending on version)

is normally separate from the one used to transfer data from a networking device to the processor. The lower-level software must know the number of buses and how to use and control them.

Let's talk about protocol stacks now. Most protocols are designed in a stacked manner, much like OSI model where highest layer on the sender's side encapsulates data in a package and lower layers encapsulate the already encapsulated data in their own respective formats. The receiver unwraps the data in the reverse order of encapsulation.